

OWNER'S SERVICE MANUAL

INSTALLATION, OPERATING & SERVICING INSTRUCTIONS



MANUAL PN 05606

**WHEELED STORED PRESSURE HALON 1211 EXTINGUISHERS
MODELS 600, 695**

All fire extinguishers shall be installed, inspected and maintained in accordance with the National Fire Protection Association standard titled "Portable Fire Extinguishers", NFPA 10 and the requirements of local authorities having jurisdiction.

When maintenance is indicated, it shall be performed by trained persons having proper equipment. Fire extinguishers are pressure vessels and must be treated with respect and handled with care. They are mechanical devices and require periodic maintenance to be sure that they are ready to operate properly and safely. Amerex strongly recommends that the maintenance of portable fire extinguishers be done by a trained professional – your local authorized Amerex Distributor.

Amerex Corporation makes original factory parts available to insure proper maintenance – USE OF SUBSTITUTE PARTS RELEASES AMEREX OF ITS WARRANTY OBLIGATIONS. Amerex parts have machined surfaces and threads that are manufactured to exacting tolerances. o-rings, hoses, nozzles, and all metal parts meet precise specifications and are subjected to multiple in-house inspections and tests for acceptability. There are substitute parts available that may be incorrectly labeled as UL component parts, some are advertised as Amerex type. None of these meet UL requirements and all of them voids the Amerex extinguisher warranty. DO NOT SUBSTITUTE.

REFERENCES IN THIS MAUNUAL:

NFPA 10 Portable Fire Extinguishers

CGA C-1 Methods for Pressure Testing Compressed Gas Cylinders

CGA C-6 Standard for Visual Inspection of Steel Compressed Gas Cylinders.

AVAILABLE FROM:

National Fire Protection Association, 1 Batterymarch Park,
Quincy, MA 02169-7471

Compressed Gas Association, 14501 George Carter Way,
Chantilly, VA 20151-2923

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**FOR REPLACEMENT PARTS SEE THE AMEREX PORTABLE AND WHEELED
PARTS BOOK PN 27277 AVAILABLE AT <http://www.amerex-fire.com> UNDER
MANUALS OF THE RESOURCE SELECTION.**

AMEREX CORPORATION DOES NOT SERVICE, MAINTAIN OR RECHARGE FIRE EXTINGUISHERS. THIS MANUAL IS PUBLISHED AS A GUIDE TO ASSIST QUALIFIED SERVICE PERSONNEL IN THE INSPECTION, MAINTENANCE AND RECHARGE OF AMEREX FIRE EXTINGUISHERS ONLY. NO INSTRUCTION MANUAL CAN ANTICIPATE ALL POSSIBLE MALFUNCTIONS THAT MAY BE ENCOUNTERED IN THE SERVICE OF FIRE EXTINGUISHERS. DUE TO THE POSSIBILITY THAT PRIOR SERVICE PERFORMED ON THIS EQUIPMENT MAY HAVE BEEN IMPROPERLY DONE, IT IS EXTREMELY IMPORTANT THAT ALL WARNINGS, CAUTIONS AND NOTES IN THIS MANUAL BE CAREFULLY OBSERVED. FAILURE TO HEED THESE INSTRUCTIONS COULD RESULT IN SERIOUS INJURY. AMEREX ASSUMES NO LIABILITY FOR SERVICE, MAINTENANCE OR RECHARGE OF FIRE

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1-Year Inspection Requirements:

NFPA 10 Maintenance is a "thorough check" of the extinguisher. It is intended to give maximum assurance that a fire extinguisher will operate effectively and safely. It includes a thorough examination for physical damage or condition to prevent its operation and any necessary repair or replacement. It will normally reveal if hydrostatic testing or internal maintenance is required.

MAINTENANCE PROCEDURE

1. Clean extinguisher to remove dirt, grease or foreign material. Check to make sure that the instruction pictogram is securely fastened and legible. Inspect the cylinder for corrosion, abrasion, dents or weld damage. If any damage is found and you doubt the integrity of the cylinder, hydrostatically test to factory test pressure, using the proof pressure method, in accordance with instructions in CGA pamphlet C-1 and C-6 and NFPA 10. See proper method of depressurizing and reclaiming Halon 1211 in Complete Maintenance procedures.

NOTE: WHEN CLEANING, AVOID USE OF SOLVENTS AROUND THE PRESSURE GAUGE. THEY COULD SERIOUSLY DAMAGE THE PLASTIC GAUGE FACE.

2. Inspect the extinguisher for damaged, missing or substitute parts. Only factory replacement parts are approved for use on Amerex fire extinguishers.

3. Weigh the extinguisher and compare with weight printed in the "Weight Block" section on the nameplate (label). Recharge extinguisher if weight is not within indicated allowable tolerances.

4. Visually inspect the pressure gauge:

- a. If bent, damaged or improper gauge, depressurize and replace.
- b. If pressure is low, check for leaks.
- c. If over pressurized (overcharged), depressurize the extinguisher and follow recharge instructions.

5. If the Model 600 is so equipped, check the agent lever indicator. The indicator arrow shall be in the Green zone. Refer to the Trouble Shooting guide if there are any problems with the indicator.

6. Check pull pin for freedom of movement. Replace if bent or if removal appears difficult.

WARNING: ALWAYS OPEN THE SHUTOFF NOZZLE HANDLE SLOWLY. ANY EVIDENCE OF AGENT IN THE NOZZLE INDICATES THAT THE UNIT MAY HAVE BEEN USED AND THE USE NOT REPORTED. BE PREPARED FOR A POSSIBLE DISCHARGE AND NOZZLE RECOIL.

7. Check the nozzle shutoff lever for freedom of movement (open and close several times). If the operation is impeded, disassemble the nozzle, replace parts and/or properly lubricate as necessary. Make sure that the nozzle tip is clear and unobstructed.

8. After making sure that there is no residual pressure in the discharge hose, disconnect it from the operating valve. Blow air through the hose and nozzle assemblies to ensure that the passage is clear of foreign material. Check the couplings, hose and hose gasket for damage or deterioration – replace as necessary.

9. Inspect the valve assembly for corrosion or damage to hose thread connection. Visually inspect the safety disc assembly on the Model 600 discharge valve. Replace valve assembly or component parts as necessary following the proper depressurization and recharge procedures. If valve removal is necessary, complete all steps in the Recharge Procedure.

10. Reconnect the hose to the agent cylinder. Properly coil the hose on the rack and install the nozzle (with the lever in a closed position) on the mount.

NOTE: When assembling the hose to the agent cylinder or nozzle to the hose, tighten the coupling 1/4 turn after contacting the hose gasket.

11. Inspect the wheels to ensure they rotate freely. Lubricate as required.
12. Check carriage assembly for loose nuts, bolts, frame distortion or damage. Check welds for damage or corrosion. Replace damaged parts or make repairs as necessary.
13. Install new tamper seal and record service data on the extinguisher inspection tag.
14. If the extinguisher has been moved to perform service, make sure that it is returned to its proper location.

6-Year Inspection Requirements:

COMPLETE MAINTENANCE (SIX YEAR MAINTENANCE)

NFPA10 Every 6 years, stored pressure extinguishers shall be emptied and subjected to the applicable maintenance procedures. When the applicable maintenance procedures are performed during periodic recharging the six-year requirement shall begin from that date.

WARNING:

- a) **Before attempting to de-valve the extinguisher for Maintenance, Hydrotest or Recharging be sure that it is completely depressurized. Halon 1211 generates a vapor pressure of 22 psi @ 70°F. NEVER VENT TO THE ATMOSPHERE. Recover agent and vapor according to the instructions below.**
- b) **Never have any part of your body over the extinguisher while removing the valve assembly.**
- c) **Halon 1211 shall not be mixed with even the slightest amount of moisture. Prolonged exposure of a de-valved cylinder to ambient air shall be avoided to prevent moisture contamination and cylinder rusting.**

COMPLETE MAINTENANCE (SIX YEAR MAINTENANCE) PROCEDURES

1. Complete items 1 through 8 in Maintenance Procedure above.
2. Attach the appropriate recharge adapter to the extinguisher operating valve on the extinguisher cylinder. Empty the extinguisher of all pressure and Halon 1211 using a listed Halon Recharge/Recovery system and a bulk Halon supply cylinder with sufficient empty capacity to accept the contents of the extinguisher.

NOTE: Every effort shall be made to halt unnecessary escape of Halon 1211 to the atmosphere to prevent detrimental environmental effect. High Efficiency Halon 1211 Recharge/Recovery (vacuum pump type) systems (UL Standard 2006) are commercially available. The Getz HR-1 (UL Approved) unit assures a minimum of 99% recovery efficiency. It allows a means of checking for and removing moisture or contamination during the recovery process.

3. When extinguisher is empty of all agent and pressure, remove valve assembly and disassemble by removing downtube, spring and valve stem assembly. Discard valve stem assembly and collar o-ring.
4. Thoroughly clean all parts of the disassembled valve with a soft bristle brush or soft cloth. Blow the valve out with dry nitrogen.

5. Install a NEW Amerex valve stem assembly after lightly lubricating the valve stem o-ring with V-711 or equivalent (do not lubricate the valve stem seal). Reassemble the spring and downtube. Carefully install a NEW collar o-ring which has been lightly lubricated with V-711 or equivalent. Set the valve assembly aside.
6. Inspect the cylinder interior following CGA Visual Inspection Standard C-6. If a hydrotest has been performed or any moisture is evident, the cylinder shall be immediately warm air dried.
7. Clean the o-ring seating groove in the cylinder neck. If any rust is evident, remove by using a fine emery cloth (200 grit). Clean the surface and lubricate the entire sealing area with a thin film of V-711 or equivalent. Install the valve assembly in extinguisher cylinder. Hand tighten firmly.
8. Use the Getz HR-1 system to purge the residual air from the extinguisher cylinder.

RECHARGE

NFPA 10 is the replacement of the extinguishing agent (also includes the expellant for this type of extinguisher.

WARNING:

- a) **Use a protective shield between you and the pressure gauge while charging an extinguisher. Do not stand in front of the gauge if a shield is not available.**
- b) **Use a regulated pressurizing source of dry nitrogen only with a minimum dew point of minus 70°F (minus 57°C). Set the regulator to no more than 25 psi above the extinguisher operating pressure.**
- c) **Check and calibrate regulator gauge at frequent intervals. The regulator gauge shall be used to determine when the intended charging pressure has been reached. Do not use the extinguisher gauge for this purpose.**
- d) **Never leave an extinguisher connected to a regulator of a high pressure source for an extended period of time. A defective regulator could cause the cylinder to rupture due to excessive pressure.**

RECHARGING PROCEDURE

1. Perform steps 1 through 8 of the "Complete Maintenance (Six Year Teardown)" section.

CAUTION: ALL EXTINGUISHER AND CHARGING SYSTEM VALVES MUST BE CLOSED BEFORE STARTING THIS PROCEDURE.

2. Thoroughly clean all parts of the disassembled valve with a soft bristle brush or soft cloth. Blow the valve out with air or nitrogen. Inspect the collar o-ring, valve stem, spring and downtube assembly, and replace parts if worn or damaged. Lubricate the collar o-ring and small o-ring on the valve stem with V-711 (do not lubricate the valve stem seal)
3. Follow all recharging instructions on Getz HR-1 or other "approved" Recharge/Recovery System.
4. Fill extinguisher with amount of halon 1211 specified on nameplate and pressurize to the pressure specified with dry nitrogen.

5. Remove the recharge adapter. Some residual halon vapor may remain in the valve orifice as a result of the charging procedure. Before attempting to leak detect, vacuum or blow the vapor away from the areas to be checked. Check extinguisher for leaks at the valve orifice, around the collar seal, cylinder welds and gauge using a Portable Halogen Leak Detector (preferred method). The alternate method is to apply leak detecting fluid or a solution of soapy water to these areas. Use dry nitrogen to blow all liquid residue out of the valve and wipe the extinguisher to dry the exterior. **DO NOT LEAVE ANY LIQUID INSIDE THE VALVE BODY.**
6. Install hose assembly, with shut-off nozzle attached to the extinguisher discharge valve. Tighten hose coupling $\frac{1}{4}$ turn after contact with hose gasket. Coil hose onto the hose rack and nozzle into mount.
7. Weigh extinguisher to confirm that the total weight is within the tolerances indicated in the Maintenance section of the extinguisher label (nameplate).
8. Record recharge date and attach new recharge tag in accordance with the requirements of the "Authority Having Jurisdiction".

